Airline Performance Analysis:

Insights on Delays, Cancellations, and Operational Efficiency

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Problem Statement

- The airline industry generates large volumes of operational data, including flight schedules, delays, cancellations, and route information.
- Frequent flight delays and cancellations impact passenger experience, operational costs, and airline reliability.
- Therefore, analyzing airline data is crucial to uncover key trends and support data-driven decision-making for improving airline and airport operations.

Objectives

- The objective of this project is to analyze large-scale airline flight data from 2019 to 2023 to uncover operational trends, delay patterns, and cancellation reasons using data visualization techniques.

- The goal is to understand airline and airport-level performance and generate actionable insights to improve operational efficiency and decision-making.

Dataset Overview

- Airline flight data covering the period from 2019 to 2023 downloaded from Kaggle.
- Includes information on flight operations, delays, cancellations, and routes.
- Contains key variables such as Airline, Departure Delay, Arrival Delay, Cancellation Status, Diversion Status and Route.
- Used to analyze operational performance, delay patterns, and seasonal trends, cancellation trends, etc.

Methodology

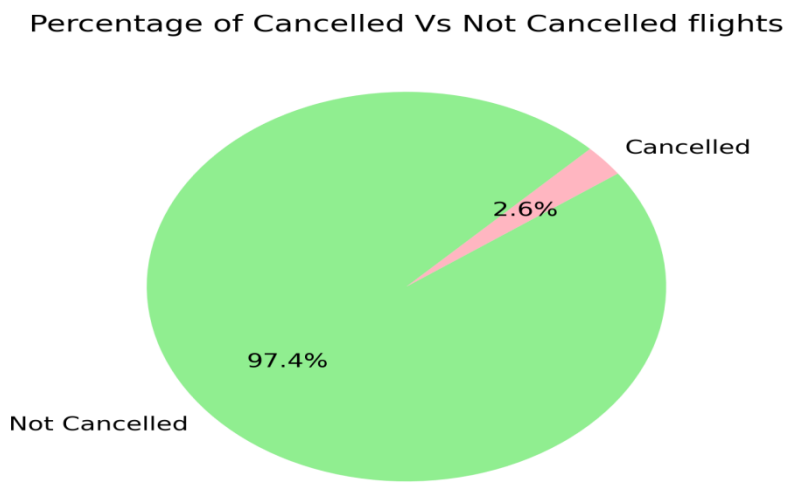
- Data Acquisition and Understanding
- Data Cleaning and Feature Engineering
- Univariate and Bivariate Analysis
- Delay Cause Analysis
- Cancellation and Seasonal Trends
- Airport and Route-Level Exploration
- Final Dashboard or Report
- Documentation and Presentation

Data Preprocessing

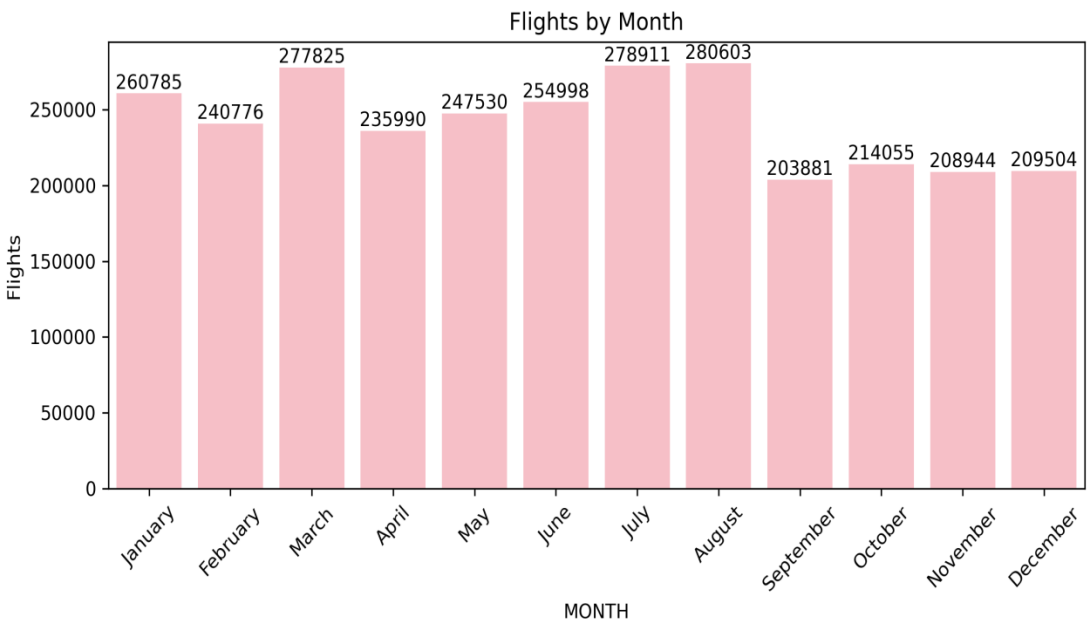
- Explored a large dataset with ~3 million records and 32 columns to understand its structure. Identified key variables related to delays, cancellations, airlines, and routes for analysis.
- Analyzed null values and found that many were due to cancelled or diverted flights. Retained these as valid nulls and replaced cancellation code nulls with “NC” (Not Cancelled).
- Removed records with missing scheduled time and inconsistent entries to maintain data quality. Verified that no duplicate records were present in the dataset.
- Converted columns into appropriate data types to improve memory usage and processing speed.
- Created new features such as Month, Day of Week, and Route for better insights. These features helped in analyzing seasonal trends and route-level performance.

Monthly Analysis

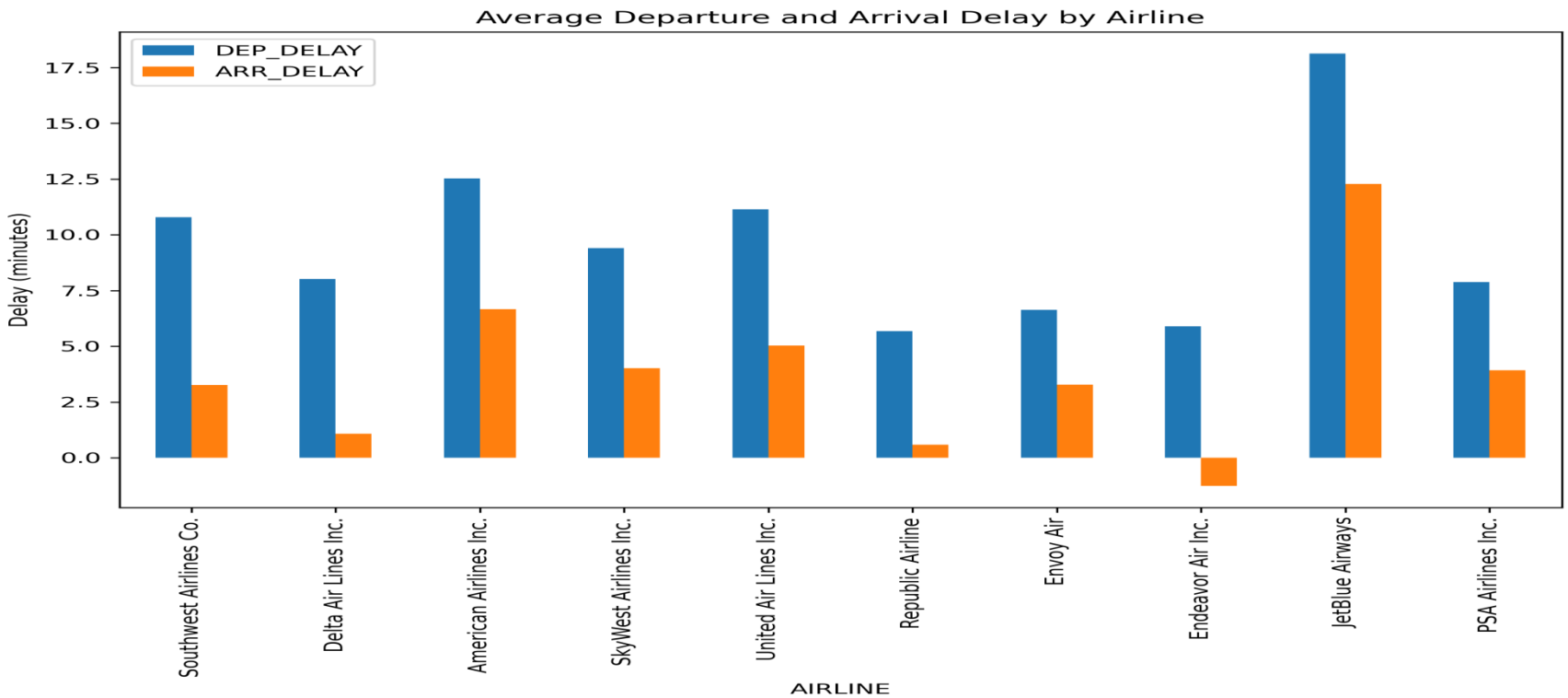
Approximately 97.4% of flights are successfully completed, while only 2.6% are cancelled, indicating strong overall operational reliability.



Flight activity peaks in August and July, indicating strong seasonal demand during mid-year travel periods. The month of Spetmeber shows less flights.

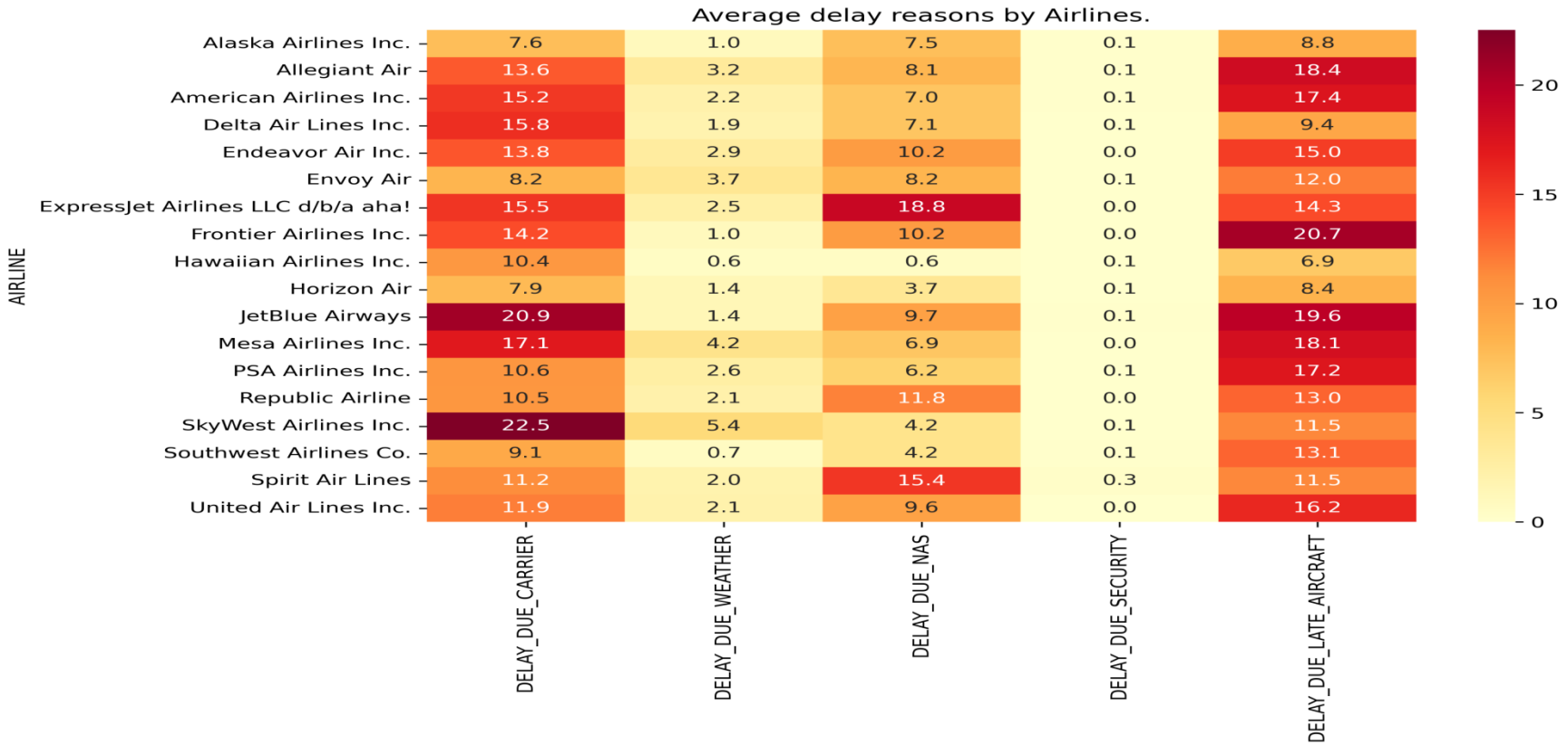


Delay Analysis



Avg DEP_DELAY is greater than ARR_DELAY acorss all the Airlines. So, flights depart late but arrive with less delay which means airlines try to make up the time during journey.

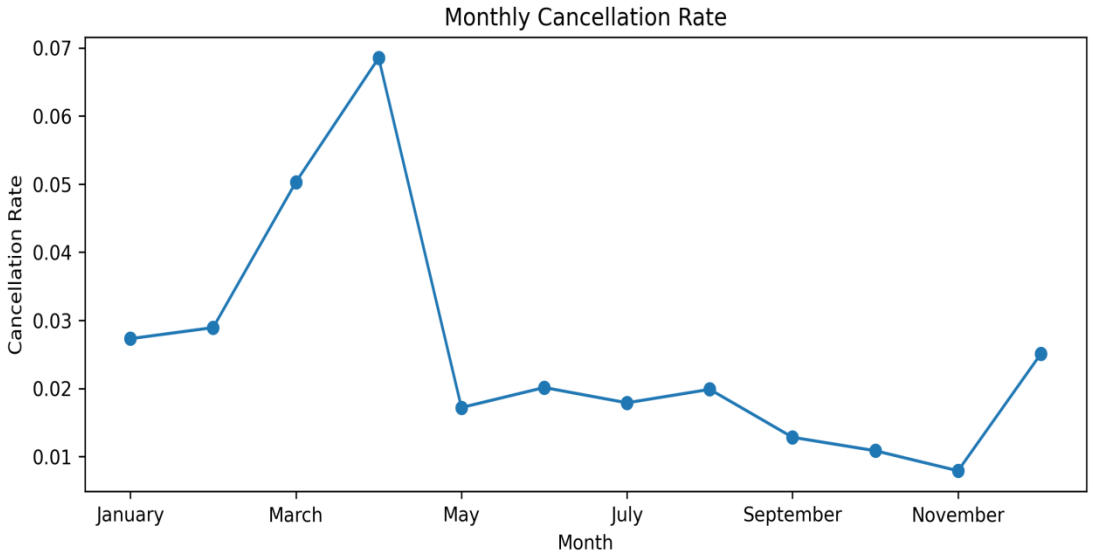
Delay Reasons Analysis



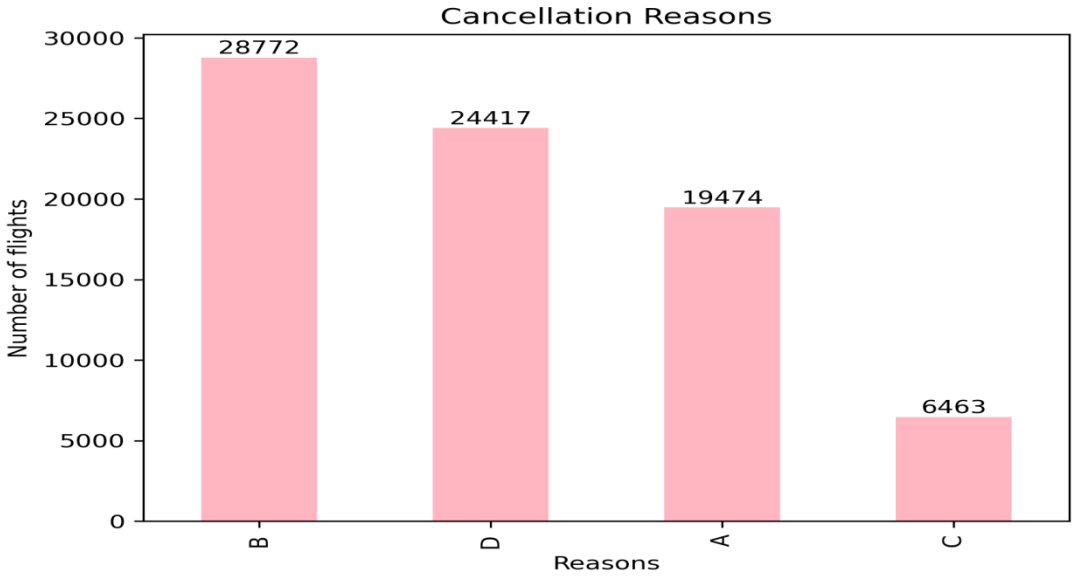
Delays are mainly driven by carrier-related issues, followed by late aircraft and NAS (air traffic congestion), indicating both operational inefficiencies and traffic-related constraints.

Cancellation Analysis

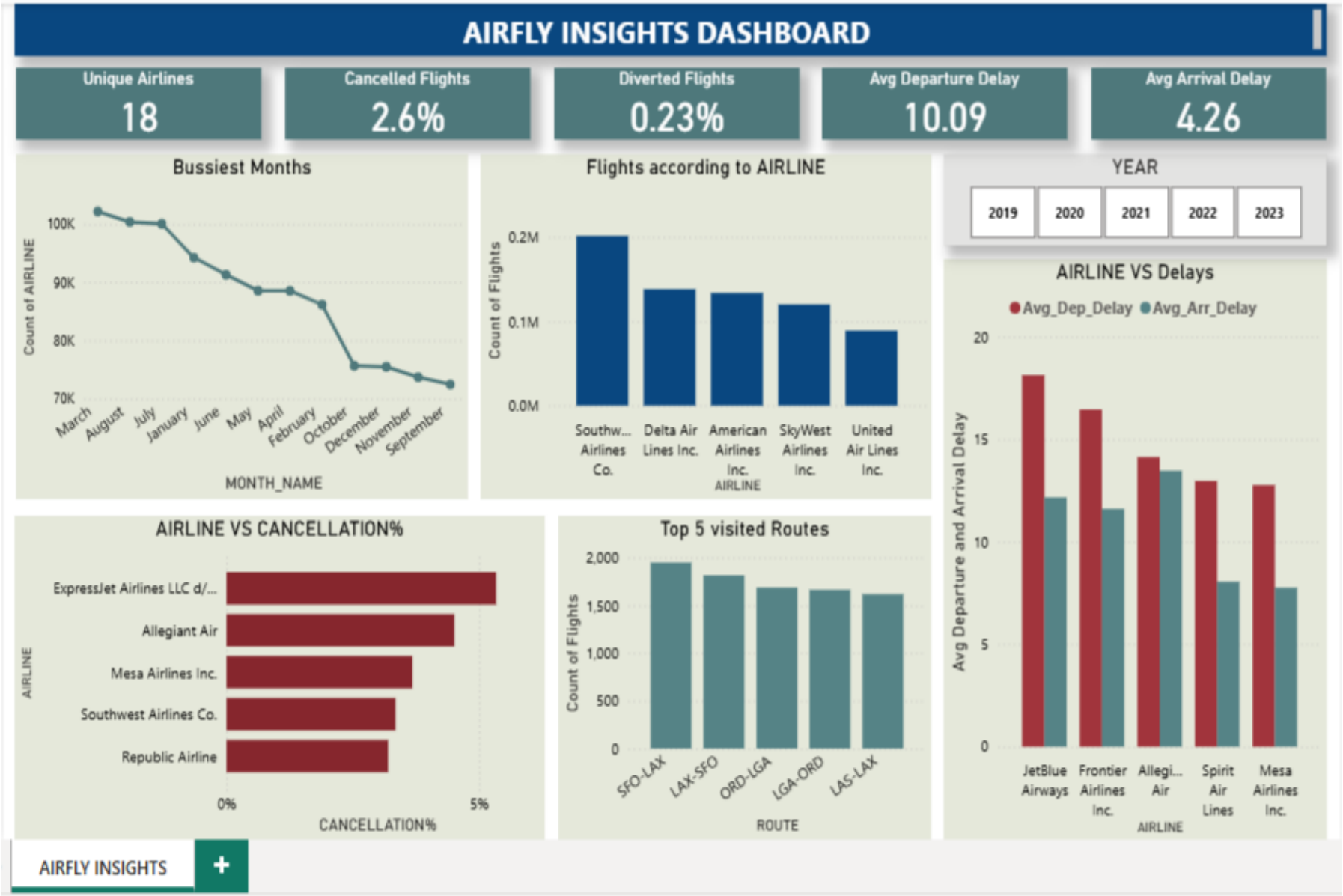
Flight cancellations peak during March and April, while November records the lowest cancellation rates. The cancellation rate does not have much impact on holiday months(November and December).



Cancellation analysis shows that most flights were cancelled due to weather and security issues. Here, A represents Carrier-related cancellation, B represents Weather, C represents NAS (air traffic system), D represents Security, and NC indicates flights that were not cancelled.



PowerBI Dashboard



Recommendations

1. Airlines should optimize internal operations such as crew scheduling, aircraft maintenance, and ground handling to reduce carrier-related delays, which are the primary cause of disruptions.
2. Since departure delays are higher than arrival delays, airlines should focus on improving turnaround time at airports and streamline boarding and departure processes to minimize initial delays.
3. Airlines should enhance resource allocation, including staffing and aircraft availability, during peak months (July, August, March) and busy weekdays to manage higher demand efficiently.
4. Given that weather is a major cause of cancellations, airlines should leverage predictive weather analytics and develop contingency plans to reduce disruptions and improve reliability.
5. Airlines should identify consistently delayed routes (e.g., FOD–DEN, DEN–ABE) and underperforming carriers to implement targeted improvements and optimize route-level performance.

Challenges Faced

- Identifying and handling null values was challenging, as many missing values were valid (e.g., due to cancelled or diverted flights) and required careful treatment to avoid incorrect analysis.
- Interpreting airline-specific variables such as delay types, cancellation codes, and operational metrics required additional domain understanding.
- Differentiating between early, on-time, and delayed flights and applying correct filtering logic was essential to ensure accurate delay calculations.

Future Scope

- The current analysis is based on historical data from 2019–2023. Future work can include real-time data to monitor and analyze flight performance dynamically.
- The analysis focuses on descriptive insights of delays and cancellations. Machine learning models can be used to predict delays and cancellations in advance.
- Current analysis identifies high-delay routes and airlines. Future improvements can focus on optimization models for better scheduling and route planning.

Conclusion

Airline operations are generally stable, but delays remain a major challenge due to operational inefficiencies and external factors. Improving scheduling, route management, and delay handling can significantly enhance overall performance and reliability.

Thank you!